Railtrack Company Code of Practice

METHODOLOGY FOR THE DEMONSTRATION OF COMPATABILITY WITH HVI TRACK CIRCUITS

ENDORSEMENT & AUTHORISATION

Endorsed by:

....

A Simmons, Professional Head of Signal Engineering

Accepted for issue by:

....

P McAleer, Acting Head of Business Standards

Authorised by:

....

A McNaughton, Chief Engineer

ISSUE RECORD

ISSUE	DATE	COMMENTS
1	February 2003	New Code of Practice

IMPLEMENTATION

The provisions of this code of practice may be used from the date of issue.

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NW1 2EE

Contents

1 INTRODUCTION

1.1 Purpose

The purpose of this document is to provide a methodology to demonstrate compatibility of trains with HVI track circuits on the a.c. and d.c. railways on Railtrack controlled infrastructure. This is achieved by:

- Providing a model of the infrastructure for the a.c. and d.c. railways
- Providing a methodology for taking the line current recordings and using the infrastructure model producing a corresponding rail to rail voltage at the receiver
- Defining a methodology to play the rail to rail voltage into a HVI receiver
- Defining the criteria for the demonstration of compatibility

These elements form the basis of a test for checking the compatibility of trains with HVI track circuits.

1.2 Scope

This document describes a method by which a train's compatibility with HVI track circuits used on Railtrack's a.c. and d.c. electrified infrastructure can be assessed.

1.3 Installation Types Covered By This Document

This document covers HVI track circuits installed in single rail mode on the a.c. and d.c. railway only.

1.4 Equipment types included in this document

The receiver equipment type covered by this document is NCO.RTV.600 which is used by Railtrack, this variant does not include the 4 terminal capacitor included in some HVI variants. Some HVI track circuits have been fitted with an external capacitor (ref.5). These will be replaced in due course by a variant specific to Railtrack in which the capacitor will be an integral part. The effect of the capacitor is to reduce the track circuits susceptibility to high frequency interference (>4kHz ref.3). By ignoring the capacitor, this methodology is pessimistic.

2. SOURCE DOCUMENTS

2.1 Group Standards

1. GK/RC0752 General information on track circuits, Issue 2, December 1998

2. GK/RT0252 Minimum requirements for the bonding of track circuits, Issue 2, February 1996.

3. GK/RC0756 HVI Track Circuits, Issue 2, December 1998.

4. Signalling Installation Handbook, GS/IH0001, Issue 2, Railtrack S&SD, January 1994.

5. RT/SIN/067 Railtrack Line Special Instruction Notice 'HVI track circuits on a.c. electrified lines', Issue 2, December 1999.

2.2 Previous Safety Case Documentation

6. PRJ-SCSE(JUN)-RP-15 GEC Juniper Project Assessment of HVI Track Circuits Response to Transient Interference (AC and DC Routes), Issue 3, October 1999

7. SAO_REF-HVI-RP-02 Juniper Project Class 334 – HVI Track Circuit Response to Transient Waveforms, Issue 2, March 2000

8. "Class 92 Locomotive EMC Safety Assessment – Volume 9, HVI Track Circuits, Fleet Operation on the Route between Crewe an Doncaster", AEA Technology ref. 120-dr0460, version b0, dated April 2000.

2.3 Other Documentation

9. High Voltage Impulse Track Circuit Receiver Sensitivity Tests, MTR 175, P. D. Booth, D. Poole, Alstom Signalling, June 1994.

10. Immunity of High Voltage Impulse Track circuits to Class 373 TMTS, MER181, D. Poole, J. Waller, Alstom Signalling, November 1993.

11. Fax from AEA Technology Rail to P. Hopkins, Railtrack, reference fx2276 dated 27th February 1997 (in support of the Class 92 locomotive EMC safety case)

2.4 Railtrack IDI Documentation

12. RT/E/C/50019 Guidance on Traction & Auxiliary Monitoring , To be issued.

3. INFRASTRUCTURE ASSUMPTIONS

The infrastructure assumed during the analysis is as described below, further assumptions are listed in Appendix B:

- The infrastructure is electrified using standard a.c., d.c. or dual electrification configurations.
- Track circuits are installed in single rail mode without the use of impedance bonds.
- Maximum length of HVI track circuit on d.c. and dual electrified infrastructure is 200m (ref. 3).
- Maximum length of HVI track circuit on a.c. electrified infrastructure is 300m (ref. 3).
- The external capacitor is ignored in this assessment.
- The rail impedance is modelled using a multifrequency track model validated in ref. 6.
- The HVI receiver is represented by a network component model to determine the rail to rail voltage at the receiver. This model is representative of the NCO.RVT.600 type. Type numbers for the equipment forming the HVI track circuit are listed in ref. 3.
- It is assumed that the HVI receiver is fitted with two BR933 slow to pick TPRs (Track Repeat Relays) or "extra delay" SSI data (ref. 3).

4. SUSCEPTIBILITY ANALYSIS

4.1 HVI Track Circuit Description

The Alstom HVI track circuit can be used for a.c. or d.c. electrified or non electrified lines, as a jointed track circuit. The track circuits are short in electrified areas, being up to 200 m in d.c. areas, up to 300 m in a.c. areas (ref. 3). They are therefore suitable in electrified areas where space to accommodate the tuned zone, required for jointless track circuits, is limited, such as points and crossings. They are also suitable for areas where there is poor electrical wheel/rail contact.

4.2 Operation of the HVI Track Circuit

The track circuit consists of a power supply, transmitter, receiver and track relay. The transmitter and receiver are coupled to the rail through transformers. Figure 1 shows the setup of HVI track circuit. This figure is reproduced from reference 3.

The wire from terminal C+ is a flying lead that is connected to one of the terminals numbered 1 to 6.

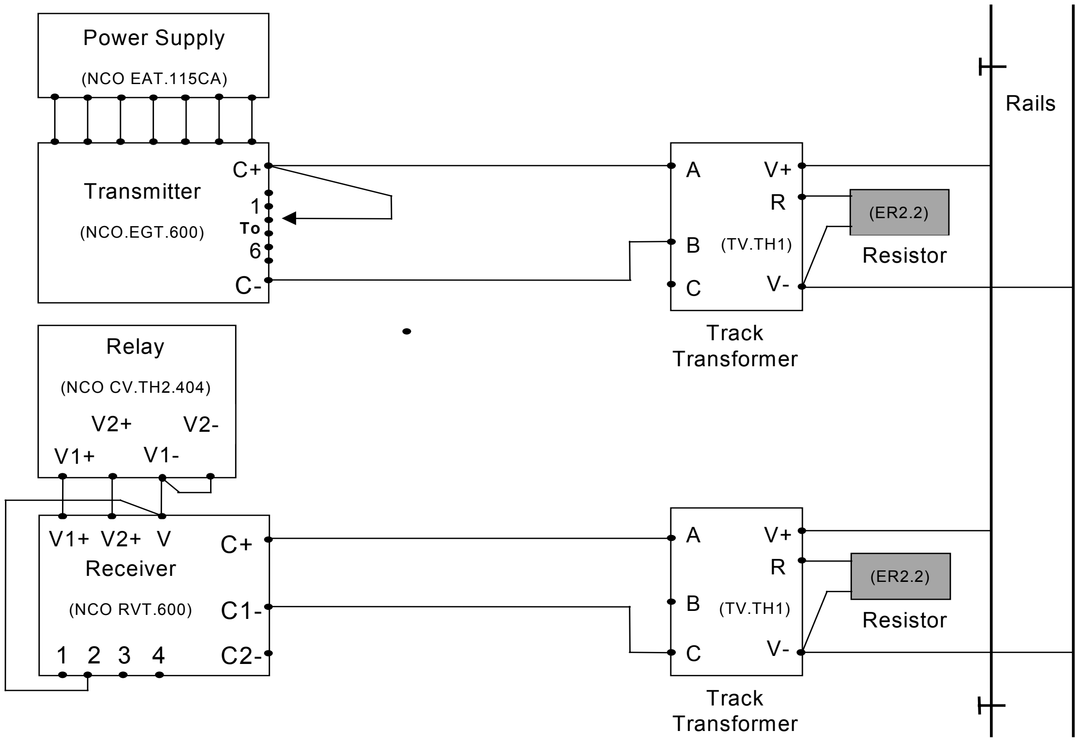


Figure 1: Setup of HVI track circuit on electrified lines

The track circuit operates by applying a short high voltage impulse to the rails, typically 100 V, three times per second. The operating pulse waveform is asymmetric with the ratio of the positive to negative voltage being 7:1. This waveform is generated by charging a capacitor to a high voltage and then discharging it through a thyristor and a track transformer. The inductance of the feed and relay end track transformers, together with that of the rails, causes the positive pulses to be followed by a negative undershoot. Figure 2 shows a typical feed end rail voltage waveform.

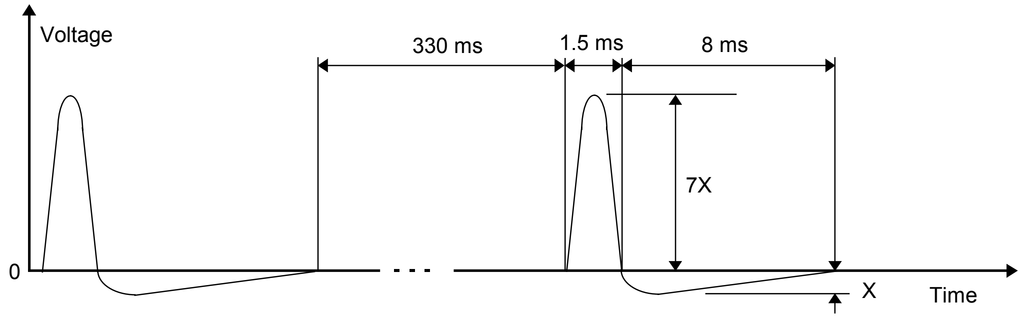


Figure 2: Typical transmitted HVI waveform

The nominal frequency of 3 pulses per second can vary and the track circuit will still work. However when the frequency is reduced to below 1 pulse per second the relay will pick momentarily and then drop (ref. 9).

At the receiver the energy from the track transformer is rectified and directly drives the track relay (see Figure 3). Therefore a power supply is not required at the relay end. The positive and negative parts of the waveform are rectified separately and applied to different windings on the relay. The flux in the centre limb of the detector cancels out when the positive and negative voltages are in the correct proportions. The flux then flows in the outer yoke picking the relay.

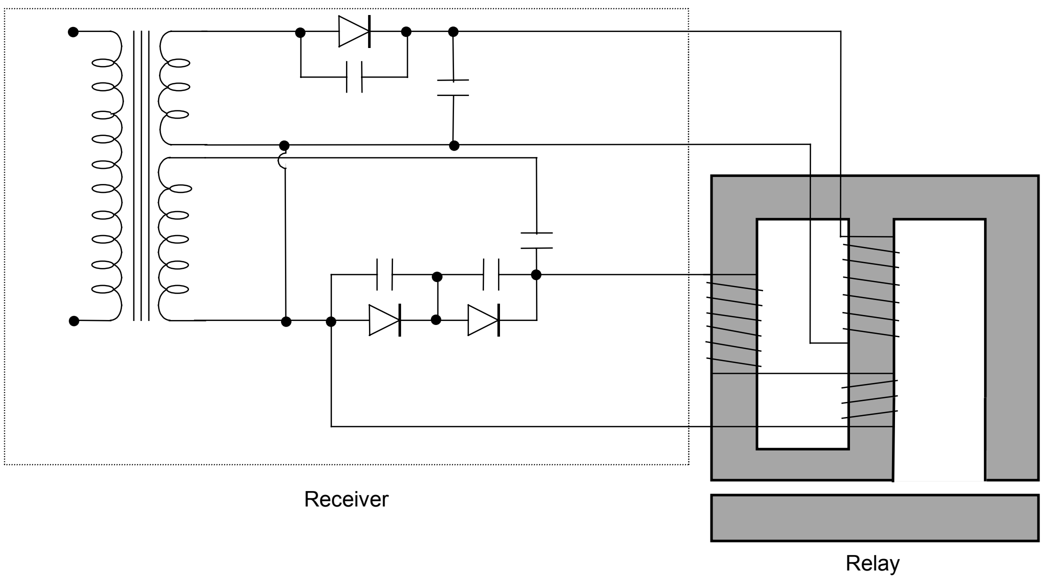


Figure 3: HVI receiver and relay

Figure 4 shows the levels of current in the relay coils required to operate the relay. The area within the defined segment represents the acceptable currents which will cause the relay to pick. If a point is plotted on this graph with the x-value representing the current in coil 1 and the y-value representing the current in coil 2, this should fall within the outlined area to make the relay pick. This characteristic is commonly referred to as the 'V-curve'.

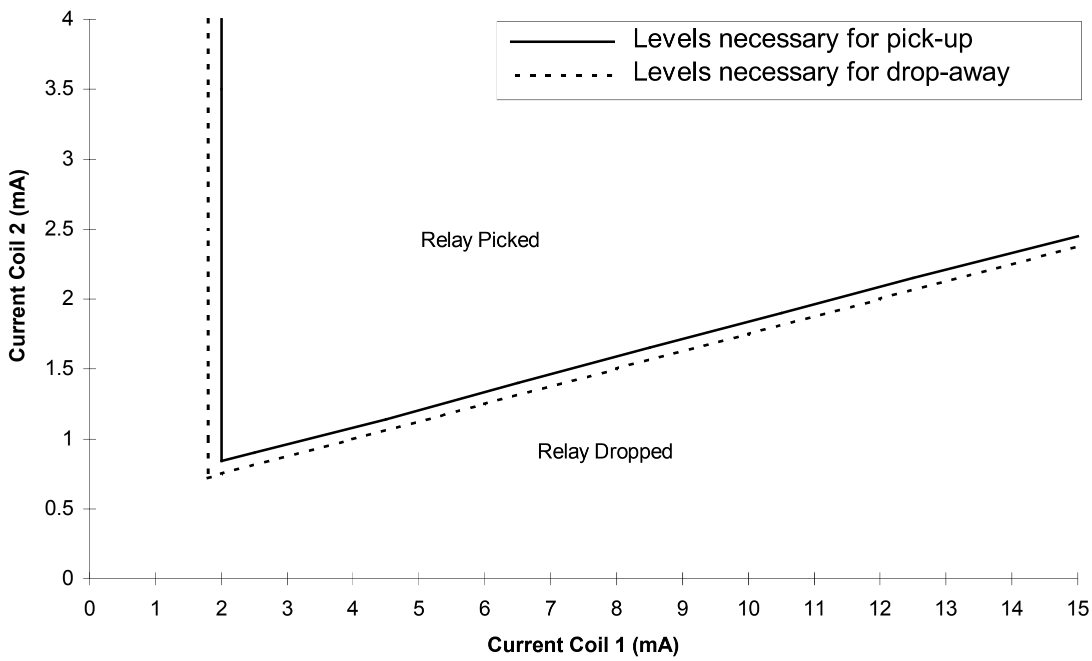


Figure 4: Levels of current in the relay coils

The drop away time of the HVI relay is 'slow' compared to the pick up time, this is the reason for the fitting of two TPRs or equivalent SSI delay (ref. 3).

4.2.1 Basic Susceptibility Criteria

Basic susceptibility criteria has been defined by the manufacturer's of the HVI track circuit in terms of the voltage developed across the receiver. This 'pick up threshold voltage' is quoted as 'about 35V' [10]. This voltage has often been converted into a rate of change of line current which develops a voltage across the traction rail which is presented across the relay. These di/dt values are 180 A/ms for a 200m track circuit (d.c. railway) and 120A/ms for a 300m track circuit (a.c. railway) [6].

4.3 Receiver Circuit Model

Note: This receiver model is used only to calculate the rail to rail voltage at the receiver it is not to be used for predicting coil currents. This model has been taken from ref. 6 and is a time domain model of the HVI receiver that can be modelled using an appropriate software package.

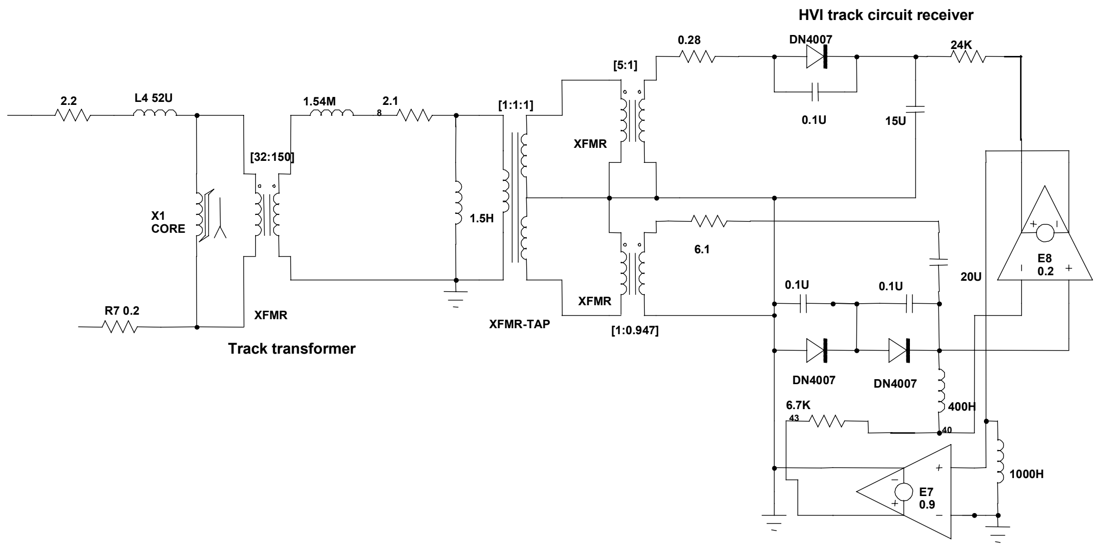


Figure 5: HVI Receiver model

N.B. The saturable core element (X1) has the following characteristics:

- Core capacity = 0.27 (Vs)
- Initial condition =0.06 (Vs)
- Magnetising Inductance = 30 (mH)
- Saturation Inductance =1.5 (mH)
- Frequency when Magnetising Reactance equals Loss Resistance = 150 (Hz)

4.4 Infrastructure models

4.4.1 Multi-Frequency Track Model

The rail impedance is a complex function of frequency which should be reflected in the time domain model to be used to determine the rail to rail voltage at the receiver. A suitable model is shown in figure 6 (ref. 6).

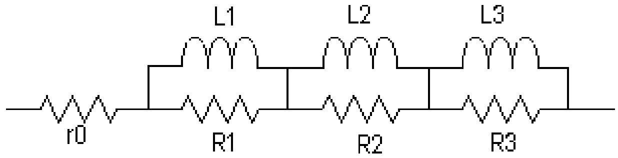


Figure 6: Multi frequency track model

The detailed solution to this model for the various infrastructure conditions can be found in references 6 and 7. These parameters are tabulated in table 1.

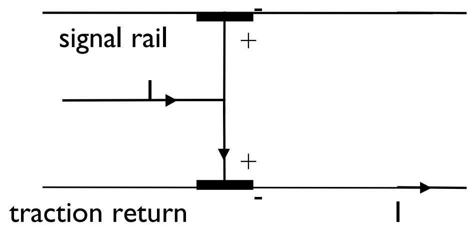
Parameter	Value for d.c. infrastructure CWR	Value for a.c. infrastructure CWR	Value a.c infrastructure Jointed rail
r0	35 mΩ/km	35 mΩ/km	250mΩ/km
R1	0.2 Ω/km	0.3 Ω/km	0.3 Ω/km
R2	5.35 Ω/km	59 mΩ/km	59 mΩ/km
R3	0.34 Ω/km	1.186 Ω/km	1.186 Ω/km
L1	3.0 mH/km	0.3 mH/km	0.3 mH/km
L2	0.3017 mH/km	0.386 mH/km	0.386 mH/km
L3	0.4273 mH/km	0.292 mH/km	0.292 mH/km

Table 1: Parameter values for Multi frequency track model for various infrastructure conditions

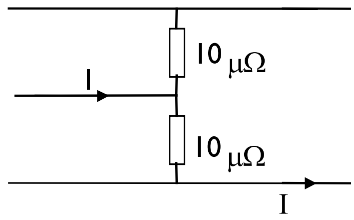
The bonds used on d.c. jointed track do not significantly affect the d.c. rail resistance and so this case is inherently included in the CWR case.

4.4.2 Shunt impedance

The shunt impedance of the train is modelled as a balanced track shunt. This is due to the nature of the transmitted voltage pulse of the track circuit any semi-conducting film that may exist at the wheel rail interface. Thus unlike other single rail track circuits a balanced track shunt model may be used.



(a) Current Flow



(b) Assumed model of train shunt

Figure 7: Single rail track circuit, current flow and assumed model of train shunt

Sensitivity analysis has demonstrated that in the case of HVI track circuits the value train shunt used for the analysis has very little effect on the predicted front contact closure time (2000% change in shunt value affected the predicted closure time by 10%, ref. 11)

4.4.3 D.C. Infrastructure Model

The d.c. infrastructure model in shown in figure 8 below. This represents a single track section of railway with a track circuit length of the maximum permitted 200m (ref. 3). CWR rail is assumed in this model. The train is represented by the current source and associated shunt resistances. The mutual impedances between the rails are included in this model.

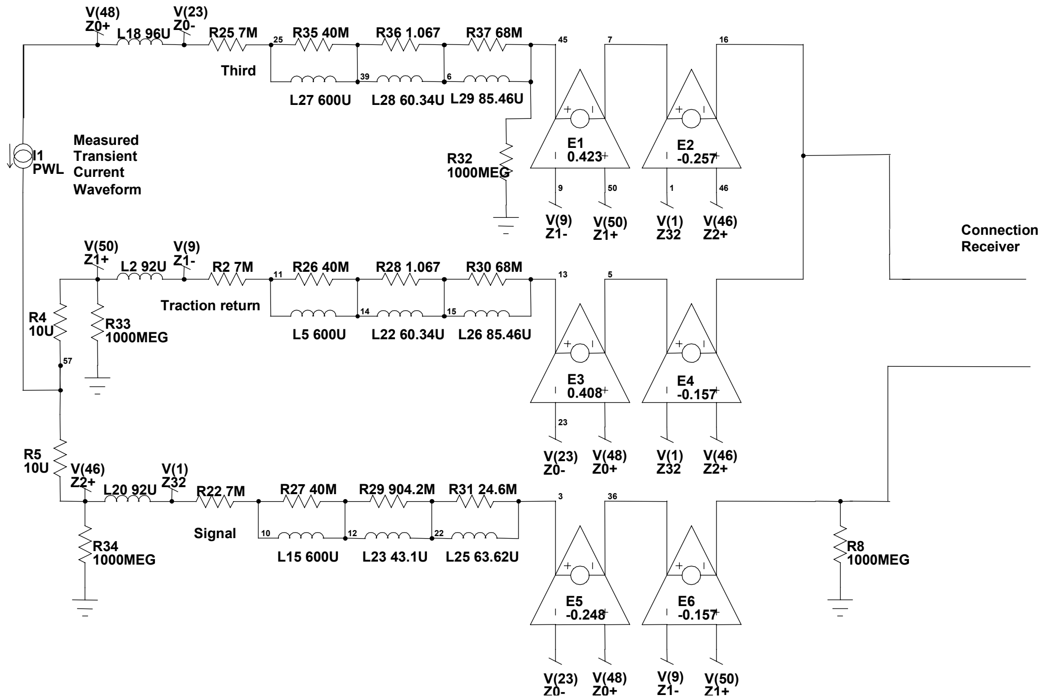


Figure 8: Computer model of a 200 m single track CWR, 3rd rail adjacent to traction return rail

4.4.4 A.C. Infrastructure Models

The a.c. infrastructure models are shown in figures 9 and 10. These are taken from reference 7.

Figure 9 shows the model for a double track section of railway with CWR. In the case shown in figure 9 the cross bond impedances have been set as open circuit to model the infrastructure in a single track configuration with intact infrastructure. The track circuit is of the maximum permitted length which is 300m (ref. 3). As in the case of the d.c. infrastructure the train is represented by the current source and associated shunt impedances. Mutual impedances between rails and overhead line are included in this model.

Figure 10 shows the infrastructure model of a 300 m long HVI track circuit on a double track section of railway with jointed rail. In this case a traction return rail is assumed to be broken.

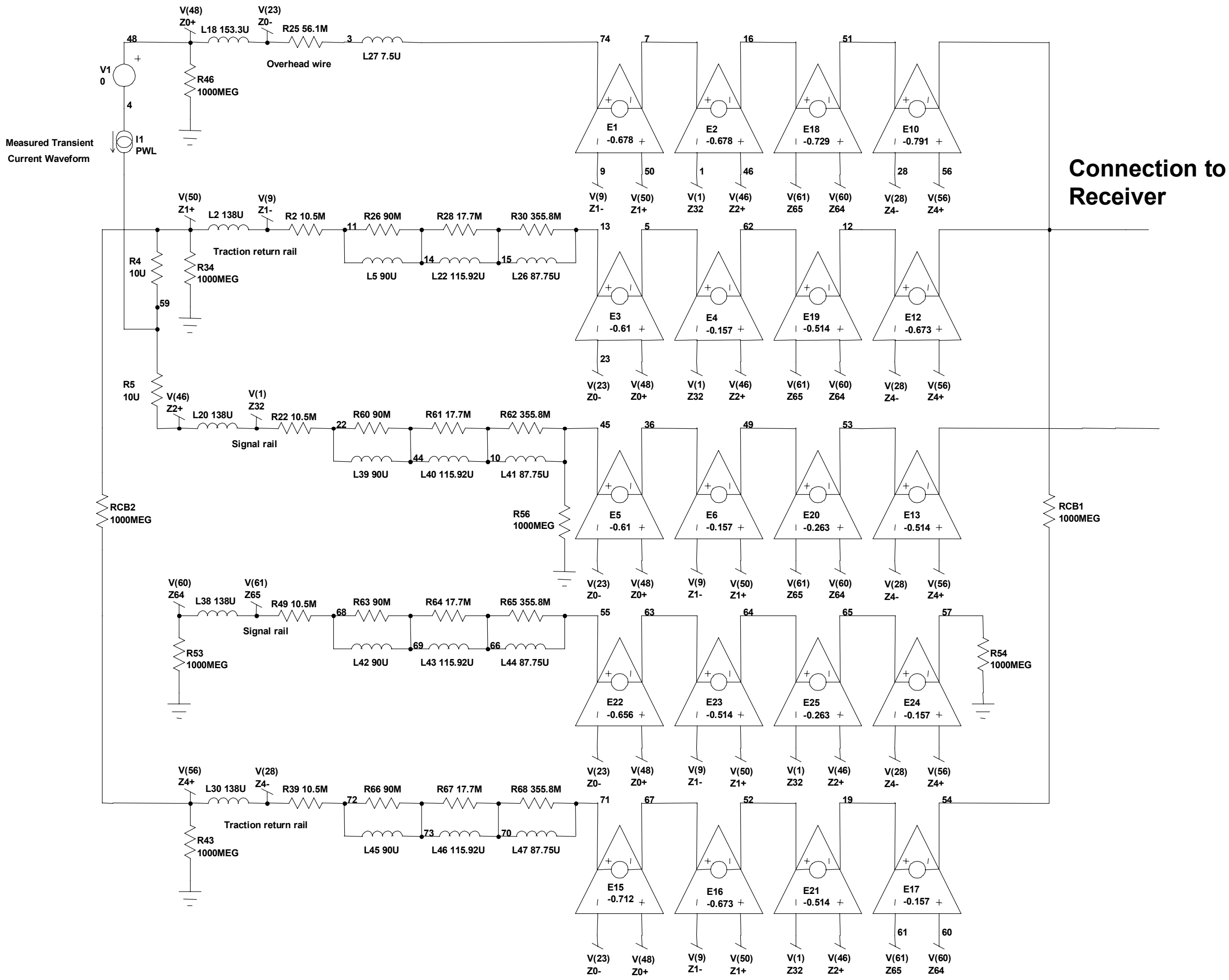


Figure 9: Computer model of a 300 m long HVI track circuit on a double track CWR with cross bonds set to high impedance to model single track – (Single track with infrastructure intact)

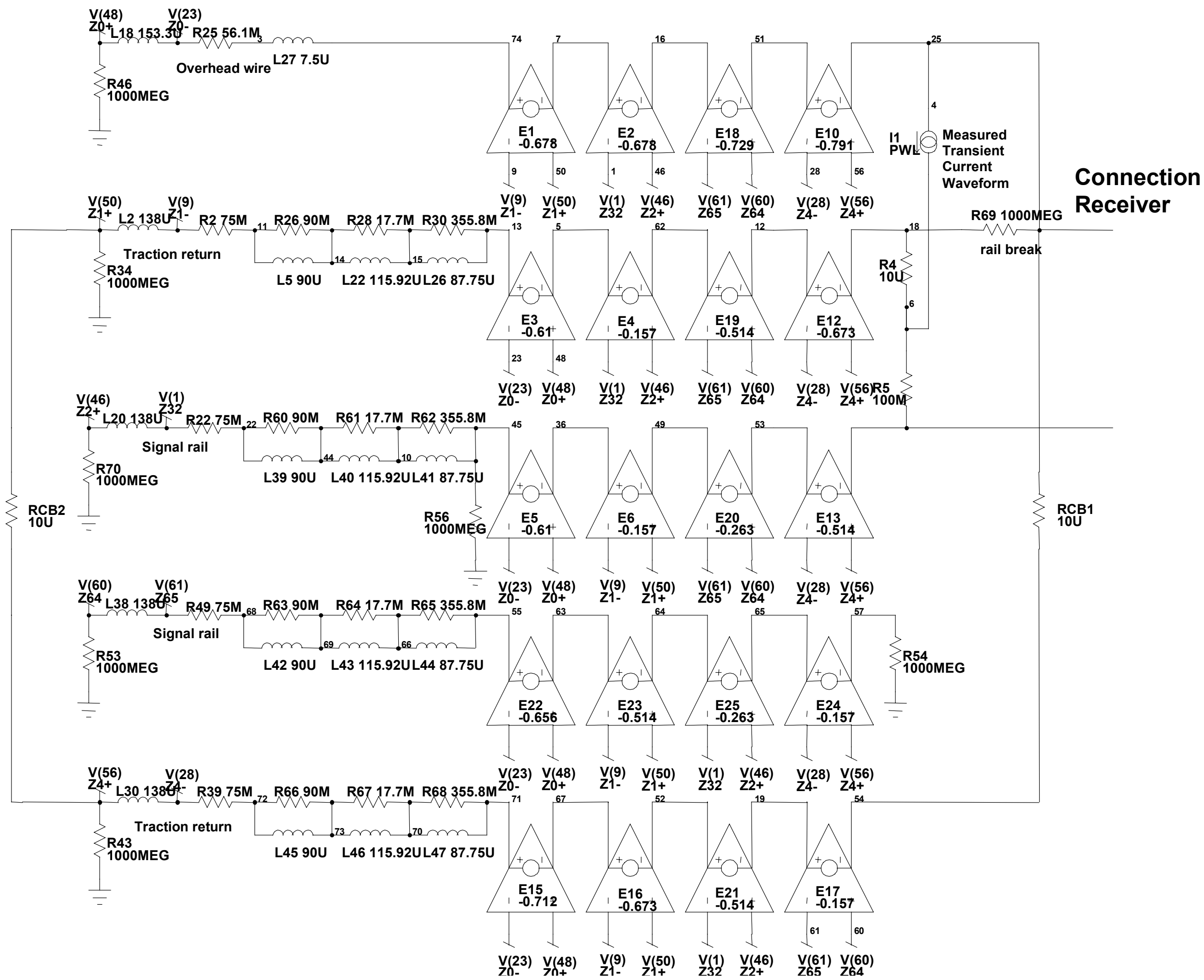


Figure 10: Computer model of a 300 m long HVI track circuit on a double track Jointed Rail – (Double track with broken traction return rail)

4.5 Susceptibility criteria

Due to the nature of the waveform used by the HVI track circuits the susceptibility criteria for compatibility cannot be expressed as a simple emission level at a particular frequency or range of frequencies. In the case of HVI track circuits the criterion is defined in terms of the time of front contact closure. This must be demonstrated to be less than the delay time provided by the rest of the signalling system. For Group Standard compliant installation this will be 800ms (ref. 3). However, there are a number of installations which do not comply with the Track Circuit Handbook; therefore the signalling system susceptibility time must be ascertained to ensure that it is always longer than the predicted contact closure time.

5. TEST METHODOLOGY

5.1 Line current recording

The detailed methodology for the measurement of line currents is given in the document 'Traction and Auxiliary Monitoring' (ref. 12). Due to the nature of the methodology described here the waveforms should be recorded in a digital format to allow them to be passed through the infrastructure model. For the purposes of demonstrating compatibility with HVI track circuits the bandwidths and therefore implied sampling rates shown in table 2 should be used:

Track circuit type	Bandwidth	Sampling rate
a.c. electrified lines (capacitor fitted)	2kHz	5kHz
a.c. electrified lines (non capacitor fitted)	8kHz	22kHz
d.c. electrified lines	2kHz	5kHz

Further detail can be found in reference 12.

5.2 Selection of waveforms

As described in section 4.2 the HVI receiver is sensitive to asymmetrical waveforms, therefore these are the types of waveforms that should be identified and played through the infrastructure model and receiver. The voltage seen by the HVI receiver is not the same shape as the line current of the traction equipment. As a first approximation the voltage across the receiver is dependent on the derivative of the line current (ref. **Error! Reference source not found.**). Therefore only line current transients with high di/dt will affect the receiver and thus need closer inspection. In addition to this events which give rise to short bursts of high frequency interference with mark/space ratios of typically 50:1 or greater.

The waveforms selected for playing through the model must include both normal and fault line current traces. 'Normal' line currents may be obtained through a programme of testing on representative infrastructure. Representative fault current traces may be obtained via testing or computer simulations of the train/traction equipment.

5.3 Selection of wave forms from testing

The data recorded during testing of a train should be analysed for events which include high levels of di/dt. The waveforms so identified should be played through the infrastructure model and receiver using the methodology described in section 5.3.

For the d.c. railway the types of events that can lead to high di/dt in the line current include but are not limited to the following:

- Gap entry
- Gap exit
- Closing/opening of the line contactor
- Transients associated with the start up of traction and/or auxiliary equipment
- Wheel slip
- Loss of line voltage
- Start up
- Line voltage steps due to other electric train operations
- Bad current collection due to ice or contamination

For the a.c. railway the types of events that can lead to high di/dt in the line current include but are not limited to the following:

- Transformer inrush on main breaker closure
- Pantograph bouncing
- Wheel slip
- Transients associated with the start up of traction and/or auxiliary equipment particularly filter charging events
- Loss of line voltage
- Shut down
- Start up
- Sympathetic inrushes caused by other vehicle inrushes
- Bad current collection due to ice or contamination
- Effect of neutral section normal and abnormal traversals

All events that give rise to step changes in line current should be checked. Due to the nature of the HVI track circuits repetitive events at relatively low frequencies (1-50Hz) require special attention.

5.4 Selection of waveforms from simulations

Simulations may be used to determine the effects of equipment failures and faults on the line current of the train. To determine the waveforms produced by train simulations that should be played through the infrastructure model and receiver the following selection criteria should be applied (ref. **Error! Reference source not found.**):

Line current di/dt >114A/ms measured between samples 800μs apart

And

Asymmetry >0.1 calculated as $ABS\left[\frac{I_{+} + I_{-}}{I_{+} - I_{-}}\right]$, where I_{+} and I_{-} are the local maximum and minimum of the waveform measured over the adjacent 100ms next to the point of maximum di/dt.

An example waveform is shown in figure 11.

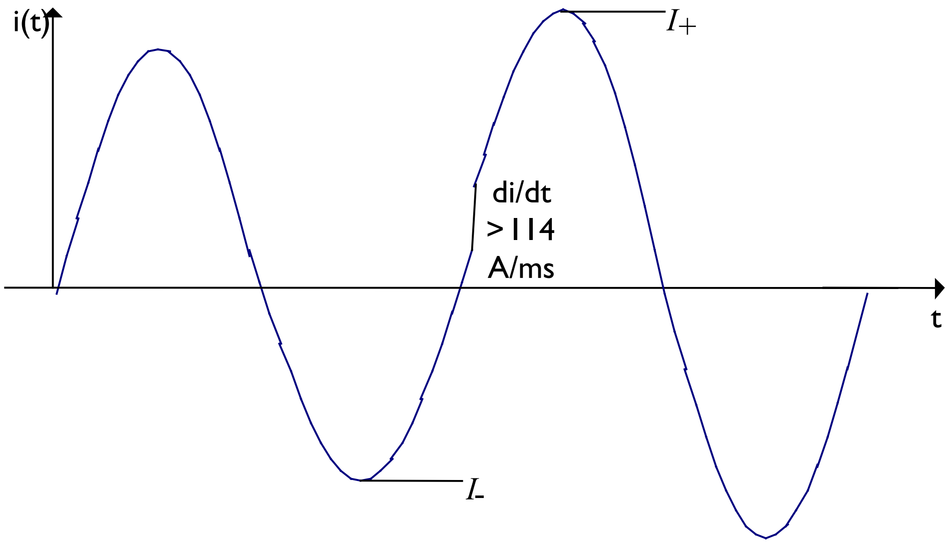


Figure 11: Example waveform

5.5 Method to play waveforms through receiver

Figure 12 shows a block diagram of the test rig to assess the response of the HVI track circuit receiver to transient waveforms (refs. 6 & 7). Figure 12.B shows the setup of equipment to monitor relay contacts. Details of the test rig equipment are given in ref. 6. The following steps summarise the procedure of how the developed test rig and track circuit models are used for the assessment of transient interference:

1. The measured transient current waveform recorded on the train is applied to the appropriate track model and the receiver voltage recorded.
2. The receiver voltage is converted into a data file containing time and voltage information.
3. The generated waveform is then applied to the actual HVI receiver via its track transformer using computer software and an arbitrary waveform generator, through a power amplifier.
4. The voltage waveform entering the track transformer, V_{RR}, and the effect of the transient interference on the relay contacts are recorded, as shown in Figure 12.
5. If the front contacts of the relay remain closed for longer than 800 ms (comparable to the combined time delay of the two repeat relays that interface with the signalling – see Section 2), the results will be interpreted as a potential wrong-side track circuit failure
6. The process must be repeated for the HVI receiver connected to the rails in both polarities as the V+ and V- terminals may be connected to either traction or signal rail to stagger the track circuits (ref. 3). The different configurations can result in different front contact closure times for the same line current recording, the longest front contact closure times must be taken in each case.

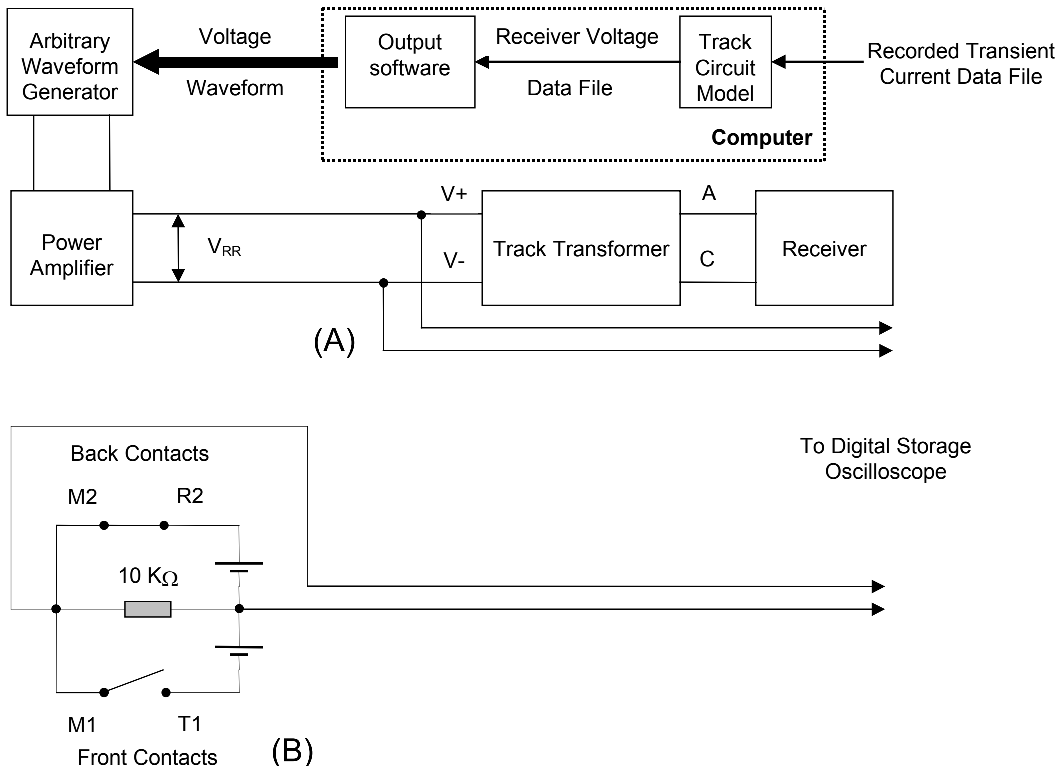


Figure 12: Setup for HVI interference tests

6. METHODOLOGY FOR DEMONSTRATION OF COMPATIBILITY

Firstly, it must be verified that the infrastructure complies with the assumptions listed in section 3. The longest track circuit on the route of interest must be determined by recourse to the RAR (Railtrack Asset Register) or other source. It is expected that this will be shorter than the 200m (d.c.) or 300m (a.c.) track circuits given in reference 3. The presence of jointed rail or CWR must be confirmed in conjunction with the track circuit length for the a.c. railway installations.

It must be confirmed that the track circuits are installed in single rail mode only and that they are fitted with the stipulated 2 TPRs or equivalent delay in SSI.

6.1 D.C. Railway

Following the methodology described in section 5.3 and the models described in section 4, it should be demonstrated that under all operating conditions the front contact closure time of the HVI relay does not exceed the compatibility criteria defined in section 4.5.

6.2 A.C. Railway

The a.c. infrastructure model described in section 4 should be configured in the following conditions:

1. Single track, CWR with infrastructure intact
2. Single track, jointed rail with infrastructure intact
3. Double track, CWR with broken traction return rail
4. Double track, jointed rail with broken traction return rail

Following the methodology described in section 5.3, under all of these infrastructure scenarios it should be demonstrated that the front contact closure times of the HVI receiver do not exceed the compatibility criteria defined in section 4.5 for all train operating conditions.

APPENDIX A: PARAMETER VALUES

Parameter values used - infrastructure model

PARAMETER	VALUES USED	REFERENCE
Resistance of all rails	Frequency dependent	See section 4.4.1
Resistance of contact wire (assuming 50% wear)	0.339 mΩ/m	Reference 7.
Resistance of catenary wire	0.415 mΩ/m	Reference 7.
Internal self inductance of all rails	Frequency dependent	Reference 7.
Internal self inductance of the contact wire	0.05 μH/m	Reference 7.
Ballast resistance	Infinite	Reference 7.
Track circuit length	300 m a.c lines	Reference 3
	200 m d.c. lines	
Cross-bond spacing	Equal to track circuit length	

All parameters are assumed to be non-variant

APPENDIX B: ASSUMPTIONS

General

A broken signal rail will lead to a protective RSF.

The infrastructure failures analysed are limited to a broken traction rail, or a broken cross bond (equating to the single track case) in a two track scenario.

Configuration

The track circuits are assumed to be installed in single rail mode with no impedance bonds present.

All of the return current flows in the rails - there is no leakage between the rails. This is a pessimistic assumption.

The effect of booster transformers is ignored i.e. rail return only. This is a pessimistic assumption.

The effects of the variation in linear track impedances due to the presence of points and crossings, check rails, metal bridges etc. are ignored. (Generally a pessimistic assumption.)

Infrastructure failure modes

It has been assumed that only one infrastructure failure can occur on one track circuit at one time.

APPENDIX C: HAZARDS

Interference current in traction return current

This hazard is covered by the assessment as presented

Axle to Axle or Longitudinal voltage

Axle to Axle voltages are not specifically considered in this document as previous assessments have indicated that they are not a threat; however, they need to be considered in every assessment.

Direct induction into track mounted or trackside components

This hazard is not covered by the assessment as presented. This has been treated by recent safety cases and the risk found to be negligible.

Particular transient events

No single type or source of transient has been identified as posing a particular risk to HVI track circuits. All transients may have the potential to affect HVI track circuits therefore train testing or simulation must include all foreseeable transient conditions.

Wheelset impedance

The maximum wheelset impedance is not quoted in this document.

Wheel/rail impedance is not significant due to the high voltage nature of the transmitted waveform.

Vehicle geometry (overhang, wheelbase etc..)

This hazard is not covered by this document.

RSF due to excessive off channel current

This hazard is not covered by the assessment as presented. This hazard has not been treated by recent safety cases. The risk presented by train generated transients that has been assessed if the false energisation of the receiver (WSF).

APPENDIX D: Mitigating Factors

Pessimistic assumptions

Zero train length for line current limits.

Rail return only for traction current.

All of the return current flows in the rails

Assumptions/variations covered by safety margin

No safety margin is applied in the case of HVI track circuits.

Background/coincident sources of interference covered by safety margin

No safety margin is applied in the case of HVI track circuits.

APPENDIX E: SOURCES OF TECHNICAL WORK

The technical work contained in this document is taken from:

1. The 'Juniper' D.C. and A.C. EMC safety cases produced by AEA Technology Rail and Alstom, ISA was WS Atkins Rail.

2. The Class 92 EMC safety case, produced by AEA Technology Rail, ISAs were WS Atkins Rail (secretariat reviewer) and Mott MacDonald.

RAILTRACK

BRIEFING NOTE

RT/E/C/50001 to RT/E/C/50019 **Issue 1**

Methodology for the demonstration of electrical compatibility between rolling stock and infrastructure

Issue Date: February 2003

Background / Synopsis

This suite of Codes of Practice is intended to assist train-builders, TOCs, and ROSCOs in the production of Route Acceptance Safety Cases for electric trains, by describing a methodology for demonstrating compatibility with EE&CS systems on Railtrack's electrified routes.

The electrical susceptibility of the key items of infrastructure is described and characterised, and recommended methods for demonstrating compatibility between rolling stock and infrastructure are given.

Key Changes / Compliance

This is a new suite of documents. As the documents are Codes of Practice, compliance is not mandatory.

Implementation

The information contained in the documents may be used with immediate effect. The various documents in the suite will be published progressively during 2003.

Of Interest To:

Authors of Route Acceptance Safety Cases.

Name and Title of Author

David Wilkinson
Technical & Safety Advisor